

Mohamed Taha Abo Heiba

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PROFESSIONAL SUMMARY

Data Engineer who designs for the failure case, not just the happy path: schema contracts at ingestion, validated history through SCD2, dimensional models (star, galaxy) built to answer questions no one has asked yet. Shipped across **Databricks, DBT, Airflow, SQL Server, and AWS. lakehouses, warehouses**, and a production RAG. **AWS Certified Cloud Practitioner and Huawei HCIA-Big Data certified**, awarded best-performing trainee.

EXPERIENCE & TRAINING

- **AWS Cloud Engineer Trainee | National Telecommunication Institute (NTI)** *Oct 2025 – Dec 2025*
 - Architected multi-tier **AWS** app (EC2/RDS/VPC/S3) using **Well-Architected Framework** for fault-tolerant, HA design
 - Ensured high availability via **RDS Multi-AZ** configuration; reduced cold storage costs by implementing S3 lifecycle policies to Glacier
- **Big Data Engineer Trainee | National Telecommunication Institute (NTI)** *Jul 2025 – Sep 2025*
 - Awarded **best-performing trainee**; earned exam voucher and passed Huawei HCIA-Big Data certification on first attempt
 - Built GB-scale **Spark batch pipelines** on Hadoop (HDFS/YARN/Hive) and **Kafka+Flink streaming pipelines** across 15+ labs, operating distributed data systems at scale across **batch and real-time processing patterns**
- **Data Engineer Trainee | Digital Egypt Pioneers Initiative (DEPI)** *Apr 2024 – Oct 2024*
 - Automated **ETL workflows** by building 4 **SSIS packages** ingesting 7 CSV sources into **SQL Server via 17 stored procedures**, reducing manual effort and accelerating data delivery
 - Delivered 70K-record dataset powering churn model at **97.4% accuracy**, enabling targeted at-risk customer retention

PROJECTS

- **Retail Lakehouse Pipeline [GitHub](#)** | PostgreSQL, Databricks, DBT, Airflow, S3, Docker
 - Built end-to-end **lakehouse: Postgres to Databricks** Bronze incremental ingestion via **Lakeflow Connect, AWS S3** as an external location for secondary ingestion, **dbt**-modeled Silver/Gold layers, **Airflow** (on **Docker**) orchestrating the full chain
 - Modeled Gold as a **Galaxy Schema** with point-in-time **SCD2** resolution across 17 dbt models and 72 passing tests
- **E-Commerce Lakehouse Databricks [GitHub](#)** | Databricks, PySpark, Unity Catalog, Jobs
 - Built an end-to-end **cloud data lakehouse pipeline** processing 200K+ records from 6 data sources through a 3-task Databricks Jobs DAG (**Bronze** ingestion → **Silver** transformation → **Gold** star schema)
 - Delivered a governed, analytics-ready Delta Lake platform using **Medallion Architecture** and **Unity Catalog**, enabling downstream **OLAP consumption** via dimensional **star schema** modeling
- **SQL Data Warehouse Project [GitHub](#)** | SQL Server, T-SQL, Star Schema, Medallion Architecture
 - Architected a fully automated **warehouse pipeline** ingesting 60K+ records **across 6 sources via 2 master stored procedures**, applying **OLTP→OLAP** separation, **star schema** design, and idempotent **ETL** execution
 - Delivered **star schema** as SQL views with **per-step execution-time logging** for direct bottleneck identification
- **EduMate RAG [Live](#) | [GitHub](#)** | FastAPI, LangChain, Qdrant Cloud, ChromaDB, Huggingface Space
 - Independently built EduMate-RAG, a decoupled microservice (**platform-integrated + standalone modes**) with **hybrid search** over Qdrant Cloud in production and used ChromaDB for dev, indexing 24K+ document chunks via automated PDF pipeline; deployed on Hugging Face Spaces at ~240ms avg latency

CERTIFICATIONS

Huawei HCIA Big Data Associate | Huawei Technologies Co., Ltd

Issued: Mar 2026 | Valid: Mar 2029

AWS Certified Cloud Practitioner (CLF-C02) | Amazon Web Services

Issued: May 2026 | Valid: May 2029

TECHNICAL SKILLS

- **Data Engineering & ETL Pipelines:** Medallion Architecture, Data Ingestion, (Batch, Stream Processing), Apache Airflow, ETL/ELT
- **Big Data & Distributed Computing:** Spark, Databricks, Lakeflow Connect, Hadoop (HDFS, YARN, Hive), Kafka
- **Data Warehousing:** Dimensional Modeling (Star / Snowflake / Galaxy Schema), T-SQL, Unity Catalog
- **Databases:** SQL Server, PostgreSQL, MySQL, HBase
- **Programming & DevOps:** Python, LangChain, FastAPI, Docker, Git / GitHub
- **Cloud & AWS Data Services:** AWS (S3, EC2, RDS, VPC, IAM, Lambda, Glue, Redshift, Athena, EMR, ELB, Auto Scaling)
- **Analytics & Visualization:** Power BI, DAX, Power Query

EDUCATION

B.Sc. Computer Science — Faculty of Computers & Artificial Intelligence, Fayoum University

Oct 2022 – Jun 2026

GPA: 3.42 / 4.0 | **Core Coursework:** Big Data Engineering, Cloud Computing, Database Management Systems, Advanced SQL, OOP, Data Structures & Algorithms, Machine Learning, Operating Systems (Linux, Docker), Python Programming, Data Visualization (Power BI)